performance concrete which shall be limited to a maximum temperature of 25°C.

415.38 Handling and Placing Concrete – In preparation for the placing of concrete, all construction debris and extraneous matter shall be removed from the interior of forms. Struts, stays and braces inside the forms shall be kept to a minimum and entirely removed when the concrete placing has reached an elevation rendering their service unnecessary.

Concrete shall be handled and placed in accordance with the requirements of CSA A23.1 and shall be placed in the shortest possible time after mixing is completed. When rectangular void forms are used, concrete under voids shall be placed and compacted before void forms are placed in position.

Concrete may be deposited a maximum of 2 m without the use of pipes, provided, when required, suitable measures are taken to prevent segregation and premature coating of upper reinforcing steel. When pipes are used, they shall, as far as practicable, be kept full of concrete during placing and their lower ends shall be kept buried in the newly placed concrete.

After initial set of the concrete, the forms shall not be jarred, and no stress shall be placed on projecting reinforcing bars.

415.39 Vibration – Concrete, during and immediately after depositing, shall be thoroughly compacted. The compaction shall be accomplished by mechanical vibration subject to the following provisions:

(a) Vibration shall be internal for all accessible parts and shall be external for inaccessible parts.

(b) Vibrators shall be of a type and design acceptable to the Ministry. They shall be capable of transmitting vibration to the concrete at frequencies recommended in CSA A23.1.

(c) The intensity of vibration shall be such as to visibly affect a mass of concrete of 50 mm slump over a radius of at least 500 mm.

(d) The Contractor shall provide a sufficient number of vibrators to properly compact each batch immediately after concrete is placed in the forms.

(e) Vibrators shall be manipulated so as to thoroughly work the concrete around the reinforcing and embedded fixtures and into the corners and angles of the forms.

(f) Vibration shall be applied at the point of deposit and in the area of freshly deposited concrete. The vibrators shall be inserted into and withdrawn from the concrete slowly. The vibration shall be of sufficient duration and intensity to thoroughly compact the concrete, but shall not be continued so as to cause segregation.

(g) Vibration shall not be continued at any one point to the extent that localized areas of grout are formed.

(h) Application of vibrators shall be at points uniformly spaced and not farther apart than twice the radius over which the vibration is visibly effective.

(i) Vibration shall not be applied directly or through the reinforcement to sections or layers of concrete which have hardened to the degree that the concrete ceases to be plastic under vibration. It shall not be used to make concrete flow in the forms over distances so great as to cause segregation, and vibrators shall not be used to transport concrete in the forms.

(j) Vibration shall be supplemented by such spading as is necessary to ensure a smooth surface and dense concrete along form surfaces and in corners and locations impossible to reach with the vibrators.

(k) Vibrator heads shall be smaller than the minimum clear distance between prestressing strands.

415.40 Test Cylinders – The Contractor shall make available, as may be required by the Ministry Representative, samples of the concrete used in casting the members.

Further, the Contractor shall take samples and make test cylinders in the presence of the Ministry Representative and in accordance with CSA A23.2-1C, -3C and -9C. At least 8 cylinders per casting bed for prestressed girders, and 4 cylinders per casting bed for non-prestressed components shall be made for testing. At the discretion of the Ministry Representative, depending on the quantity and type of element being cast, the number of cylinders may be reduced. Cylinders shall be representative of all stages of concrete placement.

415.40.01 Release Strength – Two cylinders from each end of the casting bed shall be cured in the same manner as the members and tested by the Contractor under the observation of the Ministry Representative at the Contractor's plant to assess the strength of the concrete at the time of transfer of force to the concrete. Only one cylinder from each end need be tested if the average strength of the two cylinders is equal to or greater than the required release strength at transfer and no test cylinder is more than 1.4 MPa below the release strength. No transfer of force shall occur without the acceptance of the Ministry Representative. At the discretion of the Ministry Representative, depending on the quantity and type of element being cast, the number of cylinders may be reduced.

If all release cylinders are tested and the release strength has not been attained, further testing with an impact hammer shall be allowed, provided a written procedure acceptable to the Ministry Representative is provided by the Contractor. The procedure shall correlate readings from areas of a prestressed girder where the release strength has been determined by concrete test cylinders with similar areas on

the girder in question. If this situation occurs, extra pairs of release cylinders shall be made on future casts until confidence in determining the release strength is re-established.

When HSAs are incorporated into the mix design, pre-construction tests shall be conducted to determine the rate of early strength development for the selected concrete mix design. Tests shall be repeated when there is a change in the HSA dosage rate.

415.40.02 28 Day Strength – Two cylinders from each end of the casting bed shall be identified as "acceptance cylinders" and shall be cured in the Contractor's curing tanks. These cylinders shall be tested by the Contractor under the discretionary observation of the Ministry Representative to determine the twenty-eight day strengths. The strength of the concrete in the member shall be taken as the average of the compressive strength of the "acceptance cylinders", however, no individual cylinder shall be more than 3.5 MPa below the specified strength and the average strength shall equal or exceed the 28 day strength.

415.40.03 Additional Cylinders – For special cases, extra cylinders shall be made and tested if required by the Ministry Representative. In the case of ready mix, and plant mixed concrete with no established records, extra cylinders, in addition to the release strength or 28 day strength cylinders, shall be made from each load.

415.41 Surface Finish – Throughout the work, special precautions shall be employed in formwork and concrete placing to produce a smooth and dense concrete. Exposed surfaces shall be as free as possible from form marks, fins, honeycombing and pock marking. Only a minimum amount of pointing will be allowed. Poor surface finish upon the removal of the forms shall be sufficient cause for the rejection of any member. All exposed edges shall be chamfered unless otherwise noted on the Drawings.

Roadway and surfaces to be bonded to shall be rough screeded by wood floating following initial strike off.

Surfaces to be bonded to shall be sandblasted or exposed with chemical retarder and water blasted to remove all laitance and foreign matter and to roughen the surface. The amplitude of the surface roughness shall be 5 mm, with the exception that for keyways in box beams and recesses at the lifting devices the surface roughness amplitude may be reduced to a minimum of approximately 1 mm. Loose sand and debris shall be washed or blown off. Water, cement or proprietary topping material shall not be added to the surface during finishing. The surface roughness amplitude at keyways in box beams and recesses at the lifting devices shall be approximately 1 mm.

Walkway surfaces shall be given a coarse transverse broomed finish following initial screeding and floating. Roadway surfaces not requiring any wearing surface shall be given a tined surface in accordance with SS 413.02.02(e). Roadways receiving a membrane shall be finished in accordance with SS 419.33 and SS 413.02.02(e).

The outer surfaces of all outside stringers shall be given a Class 2 finish, and any other surfaces with a different surface finish as indicated on the Drawings shall be finished in accordance with SS 211.17.

415.42 Curing – Precast and precast prestressed concrete members may be cured using non-accelerated curing or accelerated curing as outlined herein.

415.42.01 Non-accelerated Curing – Non-accelerated concrete shall be cured in accordance with the requirements of CSA A23.1 Section 7.8 except that the application of curing compound as a curing method is not permitted.

Curing Type 2 Additional curing in accordance with CSA A23.1 Table 19 shall apply to all components except that Type 3 Extended wet curing shall apply to the following components.

(a) Full depth and partial depth precast deck panels

(b) Non-prestressed components containing silica fume

415.42.02 Accelerated Curing

(a) Steam Curing – The concrete shall attain its initial set before application of steam (generally 2 to 4 hours after the final placement of concrete or as justified by previous experience or in accordance with ASTM C403/C403M Standard Test Method for Time of Setting of Concrete Mixtures by Penetration Resistance, all to the acceptance of the Ministry Representative.

If HSAs are used in the mix design, time to attain initial set after concrete has been placed may exceed that required for conventional concrete.

After the placement of concrete, and until the concrete has attained initial set, sufficient steam shall be applied to hold the air surrounding the member at a temperature not less than 13±C.

Steam curing shall be done within a suitable enclosure to contain the steam and minimize moisture and heat losses. The enclosure shall allow free circulation of steam around the sides and tops of members and shall not be in contact with the member.

Steam at the outlets shall be at 100% relative humidity and low pressure, to achieve 95% relative humidity in the enclosure. Application of the steam shall not be directly on the concrete or test cylinders and shall be from outlets distributed uniformly along the whole length of the member.

Temperatures within the concrete shall be measured and recorded in accordance with CSA A23.4 clause 23.2.3. Recording of temperatures shall continue until the end of the cooling-off period. If requested, temperature measurement records shall be made

available to the Ministry Representative. During application of the steam, the temperature within the concrete shall increase at a rate not to exceed 20°C per hour until an optimum temperature of 55°C is reached. In no case shall the temperature within the concrete exceed 70°C. The maximum temperature shall be held until the concrete has reached the transfer strength. In discontinuing the steam, the maximum cooling rate of concrete shall be 15°C per hour. Temperature monitoring shall continue until the concrete temperature is not more than 20°C above the ambient temperature. Stripping during the cooling-off period is permitted.

(b) Curing With Radiant Heat and Moisture – Curing with radiant heat shall be done by heating the casting area in combination with measures to protect concrete surfaces from moisture loss. Heat shall be applied by means of pipes circulating steam, hot oil or hot water or by electric heating elements. Pipes shall not be in contact with concrete, form surfaces or test cylinders.

Protection against moisture loss shall be by a continuous sprinkling to a cover of burlap, cotton matting covering the surface with tarpaulins or plastic sheeting, using a water spray, or ensuring that 100% relative humidity is present in the curing enclosure.

An auxiliary cover in addition to the moisture retaining cover shall be used above the entire bed to retain the heat. This cover shall be supported at a sufficient distance above the member to allow circulation of the heat.

The cycle of application of heat, temperature limitations, reduction of temperatures and use of recording thermometers shall be similar to the cycle specified for steam curing.

(c) Additional Curing Requirements – In addition to the accelerated curing, Type 3 Extended wet curing described in CSA A23.1 Table 19 shall apply to the following components.

(i) Full depth and partial depth precast deck components

(ii) Non-prestressed components containing silica fume

415.43 Draped Strands – Where called for on the Drawings strands shall be draped in accordance with CSA A23.4.

415.44 Transfer of Force to Concrete Members – The transfer of the forces in the strands to the concrete member(s) shall be in accordance with CSA A23.4 (section on Detensioning).

415.45 Protective Coating for Ends of Strands – After completion of concrete curing, the ends of tensile strands shall be ground flush with the concrete surface. Where the ends of the members are to be embedded in concrete, the ends of the strands shall be painted with two coats of an organic zinc rich paint acceptable to the Ministry. Where the ends of the members are not to be embedded in concrete, the ends of the members shall be covered with thixotropic epoxy to provide at least 3 mm cover for a band width of 50 mm on each side of all strands. This material is to be applied in accordance with the manufacturer's recommendations.

415.46 Tolerances – For bridge stringers, the tolerances specified in the PCI "Manual for Quality Control for Plants and Production of Structural Precast Concrete Products" shall be modified so that the length of each member, measured parallel to the longitudinal axis, to extreme corners, at 20°C, does not exceed the specified length by more than 10 mm / 10 m (25 mm maximum).

The Contractor shall employ an adequate method of checking tolerances both before and after placement of concrete. Review of the method of checking tolerances by the Ministry Representative will not limit the Contractor's responsibility for the accuracy of the members.

415.47 Repairs to Members – Defective areas should be repaired immediately following removal of forms and before detensioning of strands, except where observation of the defect is not possible before detensioning.

If a defect is deemed to be a structural repair and meets the criteria stated in CSA A23.4 Section 33.3, notification shall be provided to the Ministry Representative regarding the nature and extent of the defect. The Contactor shall follow the repair procedures described in CSA A23.4 Section 33.3 and submit the proposed repair procedure to the Ministry Representative for review and acceptance prior to any repair work being undertaken.

Holes, honeycomb spots, etc., shall be treated as for Class 2 finish in accordance with SS 211.17. The method of repair for cracks, broken corners and edges, bulges and other defects shall be submitted to the Ministry Representative for review and acceptance. In any case, the Ministry Representative/Inspector shall be notified regarding all repairs to precast and precast prestressed members.

415.48 Handling, Storing and Loading of Members – Members shall at all times be kept in their normal upright position during handling, storage and transportation. They shall be picked up by means of vertical forces or forces inclined towards the opposite ends of the members by no more than 30º from the vertical, applied to lifting devices near the ends of the members. I-beams shall be adequately stiffened during lifting and transportation.

Stored beams shall rest on unyielding level supports near the ends of the beams.

When loaded for transportation, members shall be supported so that they will not be overstressed. In general, beams with straight strands may be supported up to

one-tenth of their lengths from each end, whereas beams with draped strands should be supported at their bearings. If the Contractor believes that it is possible to safely support members further from their ends, the Contractor shall submit to the Ministry Representative, sketch(es), sealed by a registered professional engineer, showing the proposed support arrangement. Adequacy of support shall be based on the following assumptions:

(a) l00% impact on cantilevering ends, when checking stresses at supports,

(b) 50% negative impact on the whole member, when checking stress near midspan,

(c) 50% of prestress losses at time of shipping (unless circumstances dictate otherwise),

(d) allowable tensile stress = $0.5\sqrt{fc}$

(e) allowable compressive stress = 0.6fc

(f) where fc = estimated strength of concrete at time of shipping.

Such sketch(es) shall be submitted at the time of submittal of the working drawings (see SS 415.04). Members transported on supports further from their ends than outlined above, without the submittal of sketch(es) acceptable to the Ministry, will be rejected.

Piles shall be lifted in the horizontal position at the pick-up positions shown on the drawings and at no other points. When stacking for storage or for delivery, piles shall be supported at the pick up points. They shall be handled carefully, avoiding dropping and jarring and avoiding surface abrasions or other damage.

Any member damaged so as to be, in the opinion of the Ministry Representative, unfit for the use for which it is intended shall be replaced.

415.49 Placement of Concrete in Keyways – Keyways between box stringers shall be clean and shall be pre-wet and saturated dry surfaces, before being filled with concrete.

Concrete in keyways shall attain a strength of at least 15 MPa before light traffic such as one-tonne trucks are allowed on the bridge, and at least 25 MPa before heavy loads such as concrete trucks are allowed on the bridge.

415.50 Post-Tensioning – All post-tensioning shall be done in accordance with the SS 416 – "Manufacture and Installation of Post-Tensioned Concrete Members".

QUALITY CONTROL PROGRAM

415.61 Quality Control Program – These Subsections describe the general Quality Control Program required by the Ministry for any Fabricator undertaking the fabrication of permanent precast concrete or precast prestressed concrete bridge components. The Quality Control Program shall be part of the overall Quality Management Plan for the project.

The term "Quality Control" defines those activities that the Fabricator performs to conform to the Contract.

The term "Quality Assurance" defines those activities that the Ministry performs to ensure conformance to the Contract.

The term “Quality Audit” defines those activities that the Ministry performs to provide confidence that the QC and QA processes and resulting products satisfy the Contract requirements.

These Subsections contain statements of the Quality Objectives and Policies that the Ministry considers essential for successful and economical quality management. They also outline the Procedures and Documentation to implement and confirm that the Objectives are met.

The provisions of the Quality Control Program set forth in these Subsections shall apply to all precast concrete bridge components.

These Subsections require the establishment of a Quality Management Team with the responsibility for the successful and timely implementation of all necessary Quality Control and Quality Assurance activities. Some positions shown in the Suggested Organization Chart (see Figure 1) may be held by the same individual. For example, the General Manager may also be the Contract Administrative Manager and the Purchasing Manager. The Plant Superintendent may also be the Receiver and the Structural Supervisor.

415.62 Quality Objectives and Policies

415.62.01 Quality Objectives – The Quality Objectives of the Ministry cover all precast concrete bridge components produced under the Contract, as follows:

(a) Completed products shall conform fully to the governing Codes and Specifications stipulated in the Contract.

(b) The Quality Control Program shall be fully integrated into the ongoing manufacturing activities of the Fabricator.

(c) The operations of the Quality Control Program shall protect the interests of the Ministry with respect to scheduled delivery date and contracted price.

415.62.02 Quality Policy – It is Ministry policy that the interests of the Contractor and the Ministry coincide when a product meets the Quality requirements of the Contract, is delivered on time, and is produced in a cost-effective manner.

The Quality Control Program is an essential part of this policy. It is based on the fact that the best way to do any job is to do it right the first time. This applies to all stages of the design, drafting, procurement, manufacturing and testing of the product.

The Quality Control Program involves the Quality Management Team, Ministry Representative and Fabricator personnel. Their activities must be consistent with the Quality Objectives of the Ministry.

A Quality Control Manager shall be appointed with defined responsibilities in resolving quality matters and shall report to a senior management level. At each hold point, the Quality Control Manager shall:

(a) Document the successful completion of each stage as it progresses through fabrication and erection.

(b) Identify and report nonconforming components.

(c) Initiate or recommend disposition of nonconforming components.

(d) Verify corrections.

Any persons assigned to perform quality control inspections shall be other than those performing or directly supervising the work and they shall not report directly to immediate supervisors responsible for producing the work.

The Quality Control Program is not subordinate to any design, drafting, procurement, manufacturing and testing activities, it is an essential part of them.

415.63 Scope of the Quality Control Programs – This Quality Control Program governs the fabrication of precast concrete bridge components for the Work. The Suggested Organization Chart is shown in Figure 415- 1.

The General Manager shall be responsible for:

(a) adhering to this Quality Control Program in all respects.

(b) ensuring that completed structural components shall conform fully to the applicable design and fabrication requirements stipulated in the Contract.

(c) ensuring that all required documentation is produced according to this Quality Control Program.

415.64 Range of Capability – Within the Company, there shall be the necessary knowledge, skill (in-house or on a contract basis) and equipment to do the following work on precast concrete structural components:

(a) Design of lifting and erection devices not shown on drawings

(b) Preparation of shop fabrication and working drawings

(c) Preparation of bills of material

(d) Preparation of material requisitions and purchase orders

(e) Receiving, checking and storing materials for bridges

(f) Layout, form set-up, reinforcing steel, prestressing and post-tensioning

(g) Mix designs, batching, mixing, placing and curing of concrete

(h) Sampling, testing, making and breaking of concrete test cylinders

(i) Dimensional checking and verification

(j) Resolution of non-conformances

(k) Documentation of all stages of work with capability of tracing all major components

(l) Finishing, patching, storing and shipping

(m) Erection of bridge components

415.65 Drawings and Specifications

415.65.01 Chief Design Engineer Responsibilities – The Chief Design Engineer shall be responsible for:

(a) the design and location of lifting devices and the preparation of stressing sheets for prestressing or post-tensioning; and for the design of the erection procedures and any special erection equipment needed.

415.65.02 Chief Draftsperson Responsibilities – The Chief Draftsperson shall be responsible for:

(a) obtaining the latest revision of the design Drawings and specifications for the work.

(b) submitting working drawings, shop drawings and erection drawings to the Ministry for acceptance before commencement of the work.

(c) preparing material requisitions containing a full description of the material sizes, material specifications, and certifications required for conformance to the Contract.

(d) delivering the material requisitions to the Purchasing Manager in ample time to permit ordering, delivery and documentation and not delay the progress of the work.

(e) issuing requisitions for all sub-contracted drafting work and shall ensure that all conditions of the Contract are part of such sub-contracts.

415.66 Material Control – All materials for structural components shall be ordered by the Purchasing Manager in full conformance with the material requisitions provided by the Chief Draftsperson.

The Purchase Orders shall contain all information necessary to ensure that materials purchased will comply fully with the terms of the Contract. Where mill certificates and test reports are required, it shall be so stated on the Purchase Order. Instructions shall state when the certificates and reports are to be delivered to the Fabricator.

If a supplier proposes a substitute for any material, the Purchasing Manager shall refer the proposed substitution to the Chief Design Engineer for review. If the substitute is acceptable to the Ministry, the Chief Draftsperson shall

amend all drawings and requisitions, withdraw old issues and issue the new versions.

The Receiver shall:

(a) inspect all materials on arrival for conformance with the Purchase Orders.

(b) confirm that mill certificates and test reports are provided and that they correctly identify the materials delivered.

(c) arrange with the Plant Superintendent to store all materials for the Contract in segregated areas.

(d) ensure all materials are clearly identified with the Contract.

(e) issue a non-conformance report covering overage, shortage or damage to the materials. Copies of the report shall be provided to the Purchasing Manager and the Quality Control Manager.

The Purchasing Manager shall deliver all documentation to the Quality Control Manager for inclusion in the Quality Control file for the Contract.

415.67 Pre-Pour Inspection and Reporting – The Quality Control Manager shall ensure only documented materials are used for the work.

Forms used in the casting of members shall be inspected prior to use and shall meet all the requirements of CSA A23.4.

Reinforcement shall be fabricated and placed to satisfy SS 412 and CSA A23.4 and the detailed shop drawings for quantity and spacing. Any reinforcement used to anchor cast in hardware by being welded to it will require the Fabricator to be certified to CSA W186-M.

All cast in hardware shall be fabricated and placed according to CSA A23.4. Any hardware requiring welding, including stud welding, will require the fabricator to be certified under CSA W47.1 Division 2 and have approved procedures for each weld.

All pre-tensioning operations shall be done according to CSA A23.4. Care shall be taken to properly execute, supervise and record the operations in a clear and legible manner.

The Quality Control Manager shall file a written report of verification with the General Manager and report any non-conformance that may exist on any of the above items in SS 415.67.

415.68 Concrete Control – All materials used in concrete mixes such as cement, fine and coarse aggregates, and admixtures, shall have representative certificates that can verify the quality of them.

Production of concrete shall be done according to the Contract. Records shall be kept of the mix type and quantity used in each pour.

Handling, placing and vibration of freshly mixed concrete shall be done as per Contract specifications. Any variations and interruptions in the process shall be recorded.

The sampling and testing of freshly mixed concrete shall be in accordance with the specifications and procedures laid out in CSA A23.2. Concrete test cylinders shall be made, cured and tested in accordance with the Contract specifications.

Reports for all concrete testing shall be kept on file and all non-conformities reported.

415.69 Post Pour Inspection and Reporting – Dimensions of all members shall be verified to ensure tolerances are being maintained as per the Contract specifications.

Verification shall be made of finishes, blockouts, keyways, projecting rebar, sole plates and protection of the cut off strands. Camber shall be monitored at release and 28 days.

Members shall be stored according to the fabricator's documented procedures.

Members shall only be allowed to be shipped out for erection after they have been approved by both Quality Control and accepted by the Ministry quality assurance representative. A certificate of compliance may be issued by the Ministry quality assurance representative to acknowledge acceptance.

Records shall be kept on file for all items in this section.

415.70 Correction of Non-Conformities – When a non-conformance is encountered, the Quality Control Manager shall determine a recommended disposition and submit such disposition to the Ministry Representative for acceptance as quickly as possible.

If there is non-conformance to the material specification stipulated in the Purchase Order, the Purchasing Manager shall immediately find out the reasons for the delivery of non-conforming materials. If the material is of a grade superior to that ordered, the Quality Control Manager and the Chief Engineer shall be notified. They shall verify that the material is an acceptable alternative in all respects. This verification shall include consultation with the Ministry. If the material is of a grade inferior to that ordered, it shall be rejected and the correct material or a superior material shall be obtained.

If there is non-conformance of material delivery that will delay production, the Contract Administrative Manager and Plant Superintendent shall be notified immediately. They shall be given revised delivery dates for the adjustment of production scheduling. The General Manager shall determine the alternatives available and shall notify the Ministry.

If there are pre-pour non-conformities in form set-up, prestressing, rebar placement, hardware and blockout size and location, the Quality Control Manager and Structural

Supervisor shall immediately review the non-conformance and notify the Chief Design Engineer who may require further investigation prior to submitting corrective action to the Ministry for acceptance. If the necessary corrective action will result in delay to production, the General Manager shall be notified for adjustment to the production schedule. The Plant Superintendent shall inform the Quality Control Manager when the corrective actions are being done so that conformance can be verified and the non-conformance report cancelled.

If there are post-pour non-conformities such as honeycombs or large voids after form removal, before de-tensioning the Quality Control Manager shall notify the Chief Design Engineer who will investigate and submit corrective action to the Ministry for acceptance. Other non-conformities such as low concrete strength, dimensional errors due to faulty forms, variations in camber from design and missing hardware shall all be reported by the Quality Control Manager to the Ministry Representative.

415.71 Welding – All welding on structural components shall conform to the Welding Standards of the Contractor as approved by the Canadian Welding Bureau.

All welding consumables shall conform to the approved Welding Standards and shall be received, stored and conditioned according to the applicable Welding Standards.

All welders and welding operators on the Contract shall be qualified under the requirements of the current CSA W47.1 division 2 or better.

All welding of reinforcing steel, when acceptable to the Ministry, shall be done by fabricators certified under CSA W186.

415.72 Calibration of Measurement and Test Equipment – All measurement and testing equipment owned by the Fabricator and used in the Quality Control Program shall be calibrated and re-calibrated at the intervals and in the manner stipulated in CSA A23.4.

When the concrete testing services are contracted to a Qualified testing firm the testing firm shall meet the requirements of CSA A283. The testing firm shall confirm in writing to the Ministry Representative that all their measurement and test equipment used in the testing is properly calibrated. If requested, they shall furnish copies of the calibration test data for inclusion in the contract documentation.

415.73 Records Retention – The Fabricator's Record File for each Contract shall contain the pertinent drawings, purchase orders, bills of material, material mill certificates, test reports, Quality Control documents, and certificates of compliance and be made available to the Ministry's Inspectors upon request.

Items in the Fabricator's Record File shall be retained as per company policy.

The Quality Control Manager shall ensure that each file is complete in all respects before it is placed in the Company archives.

415.74 Hold Points – In planning the workflow, it is essential to coordinate with the Ministry Representative to decide "hold" points (as defined in SS 145.12) for inspection. A list of hold points shall be drawn up by the Fabricator and the Ministry at a prefabrication meeting. The Ministry should be continuously informed of progress so that delays are minimized.

The "hold" points will typically include, but not be limited to, the following:

(a) verification of form size, quality and layout

(b) after stressing, rebar and hardware placement

(c) prior to form close up

(d) sampling and testing of concrete

(e) release cylinders

(f) after patching and finishing

(g) 28 day cylinder results

(h) shipping arrangements.

It must be understood that work will not proceed past a "hold" point until it has been signed off by Quality Control and Quality Assurance. Reports must be completed promptly.

(Refer to the current edition of the PCI Manual – Publication MNL-116 for "Sample Record Forms")

MEASUREMENT

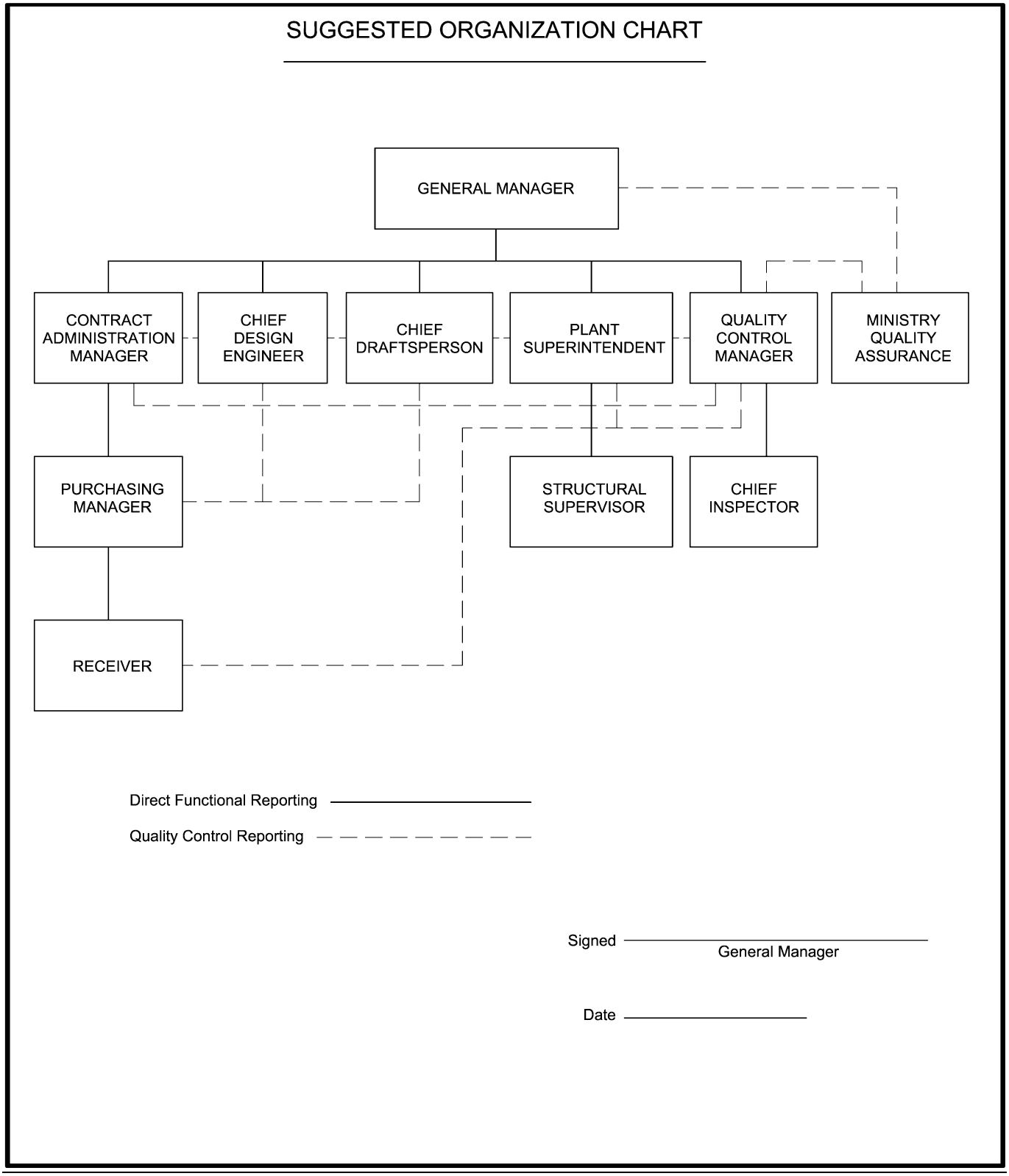
415.81 Precast Concrete Members – Precast Concrete members will be measured by EACH of the type specified.

PAYMENT

415.91 Supply and Fabrication –Payment for supply and fabrication of PRECAST CONCRETE MEMBERS will be made at the lump sum price bid. Payment shall be for quality control, working drawings, supply and placing of concrete, formwork, reinforcing steel, prestressing strands and any required inserts. Payment shall also cover tensioning, curing, release, stripping of forms, patching as necessary and storage of members.

415.92 Shipping and Erection – Payment for shipping and erection of PRECAST CONCRETE MEMBERS will be made at the lump sum price bid. Payment shall be for quality control, loading, shipping, and unloading of members. Payment shall also cover falsework and erection of members. For box stringers, payment shall also cover the supply and placement of sealing strips between stringers.

Figure 415- 1: Suggested Organization Chart



SECTION 416

MANUFACTURE AND INSTALLATION OF POST-TENSIONED CONCRETE MEMBERS

DESCRIPTION

416.01 Scope – This Section covers the manufacture and installation of post-tensioned concrete members.

416.02 General

416.02.01 General Requirements – The requirements of CSA-S6, the Ministry Supplement to CSA-S6 and SS 415 shall apply unless specified otherwise herein or in the Contract.

For items not covered in CSA-S6, the Ministry Supplement to CSA-S6 SS 415 or the Contract, then CSA A23.1, Subsection 6.8, shall apply.

416.02.02 Post-Tensioning Procedure – Three weeks before post-tensioning, the Contractor shall submit to the Ministry Representative for review, complete details of the proposed procedure, including jacking sequence and loads, procedure for testing grout cubes, and a list of equipment. This procedure shall be prepared and sealed by a professional engineer registered with the Association of Professional Engineers and Geoscientists of the Province of British Columbia (APEGBC).

MATERIALS

416.11 Anchorages – Anchorages tested in an unbonded state shall develop at least 95% of the minimum specified ultimate strength of the prestressing tendons, without exceeding anticipated set and without slippage. Test results or certificates meeting these requirements shall be supplied by the Contractor.

Anchorages shall be able to sustain the loads applied to them without slippage, distortion or other changes that will permit loss of prestress.

The anchorages shall be of such a nature that they will not kink, neck down or otherwise damage the prestressing steel.

416.12 Ducts – The ducts shall be provided with vents at all high points and at the anchorages, so as to facilitate grouting. Drains may be provided at low points. In any case, in cold weather, care shall be taken to prevent water from collecting and freezing in the ducts. The locations of the vents and drains shall be as shown on the Drawings.

Cored and external ducts shall be watertight when tested under an internal water pressure of 0.35 MPa.

The vents and drains shall be 13 mm minimum diameter flexible tubes. The connections to the ducts shall be made with metallic structural fasteners, or an alternate approved by the Ministry Representative.

416.12.01 Internal Tendons – The sheaths for post-tensioning ducts shall be made of galvanized steel or plastic. Plastic sheaths shall meet the requirements of SS 416.12.02. The sheaths shall be corrugated, and chemically non-reactive with concrete, tendons, or grout. The inside cross-sectional area of the sheaths shall be at least two times the area of the prestressing steel.

Rigid steel sheaths shall have a wall thickness of at least 0.60 mm and shall be capable of bending to an inside radius of 9 m without distress. Semi-rigid steel sheaths shall have a wall thickness of at least 0.25 mm and shall be capable of bending to an inside radius of 3.50 m without distress.

The ducts shall be spliced with telescopic sleeves. The joint lines between the ducts and the sleeves shall be wrapped with tape or heat-shrink material to make the connections mortar-tight.

416.12.02 External Tendons – Unless otherwise specified in the Contract, sheaths for external post-tensioning shall be made of plastic. Plastic sheaths, including their splices shall be made of high density polyethylene conforming to ASTM D3350, cell classification 334420C. The polyethylene sheath shall be manufactured in accordance with ASTM D2239.

Plastic sheaths shall not be used when the radius of curvature of the tendon is less than 10 m. Sheaths and their splices for external post-tensioning shall be smooth, seamless and capable of withstanding a grouting pressure of at least 1 MPa.

416.13 Condition – Tendons, ducts, anchor assemblies and other accessories shall be free of grease, oil and other foreign matter. A light coat of rust will be permissible provided any loose rust is removed and the surface of the steel is not pitted.

416.14 Epoxy – Epoxy applied to match-cast segment ends shall be as specified by AASHTO LRFD Bridge Construction Specifications , Section 8.13.7.

416.15 Grout – Cement**,** water and admixtures for grout shall conform to the requirements of CSA A23.1, Clause 4.2. Unless otherwise specified, grout for post-tensioning shall have a compressive strength of at least 20 MPa at 7 days and 35 MPa at 28 days. The Contractor shall propose a design mix for the grout and submit it for review and acceptance by the Ministry Representative before grouting may begin.

The Contractor shall supply all materials for the grout.

The grout shall have a maximum water/cement ratio of 0.45. The grout shall consist of a mixture of Type GU (General Use) or Type GUL (General Use Limestone) cement, water and an approved admixture to control expansion and

pumpability. The admixture shall be an aluminum powder base premixed material. Type HE (high early strength) cement may be substituted for Type GU or Type GUL during low temperature periods. The Contractor shall determine the dosage of the admixture by using volume change and bleeding tests conducted before grouting commences. The Contractor shall make test grout cubes and deliver them to a testing agency acceptable to the Ministry Representative.

EQUIPMENT

416.21 Stressing Equipment

416.21.01 Supervision – The Contractor shall provide a supervisor skilled in the use of the system of post-tensioning to be used.

416.21.02 Gauges and Calibration – The tensioning system shall be equipped with analog or digital gauges that have been calibrated for the specific jacks and controls by an approved testing agency within the last 6 months. The dials of the analog gauges shall be at least 150 mm in diameter. Before stressing commences, certified calibration curves shall be furnished by the Contractor.

The tensioning system or tensioning devices shall be recalibrated if during the work any gauges give erratic results.

416.22 Grout – The grout shall be mixed in a shear type (colloidal) mixer with sufficient capacity to guarantee a continuous supply of grout to the pump. The accessory equipment to batch all materials shall accurately measure all materials.

The pump shall be a positive-displacement type and shall produce an outlet pressure of at least 1 MPa. The pump shall have seals to prevent introduction of oil, air or other foreign substance into the grout and to prevent loss of grout.

The grouting equipment shall utilize gravity feed to the pump inlet from a hopper attached to and directly over it. The hopper must be kept partially full of grout to prevent air from being drawn into the ducts.

The grouting equipment shall contain a screen having clear openings of 2.5 mm maximum to screen the grout prior to its introduction into the grout pump. A screen opening of 5 mm is satisfactory if a grout with a thixotropic additive is used. This screen shall be readily accessible for inspection and cleaning.

The Contractor shall have on site a pressure gauge with a full-scale of no greater than 2 MPa. The Ministry Representative may request the Contractor to install the gauge in the grout line between the pump outlet and the duct inlets to confirm pumping pressures.

The grouting equipment shall be capable of continuously grouting the largest tendon on the project in 20 minutes.

CONSTRUCTION

416.31 Anchorages – The spacing between anchorages shall be sufficient to allow the operation of the stressing jacks to be unimpeded by the adjacent tendons.

The axis of the tendons shall be in line with the anchorages for a minimum of 400 mm.

The load from the anchoring devices shall be effectively distributed to the concrete.

Anchorages shall be rigidly held in position when the concrete in the section is placed. Anchorages shall hold the tensioned tendons without slip of more than 6 mm; otherwise detensioning will be required.

416.32 Ducts – Vents and drains in the ducts shall be mortar-tight, taped as necessary and shall provide means for injection of grout and for sealing. The ends of vents and drains shall be removed 25 mm below the concrete surfaces after grouting has been completed. If drains are used, they shall be vented downward and shall remain open until grouting commences to permit the removal of any water that may be present in the duct.

416.33 Placement and Protection of the Tendons and Other Accessories – The plan and profile location of the ducts shall be as shown on the Drawings. Ducts shall be securely held at intervals of 1000 mm or less to prevent displacement or flotation during the placement of concrete. They shall be placed with a tolerance of ±6 mm in both the vertical and the horizontal directions. The location of the ducts shall permit the post-tensioning steel to follow the prescribed profiles.

The anchorage assemblies shall be placed so that the bearing surfaces are concentric with and perpendicular to the tendons within ±1°.

Immediately before concrete is placed, the post-tensioning steel profile and alignment shall be checked and corrected where necessary. The ducts, grout fittings, vents and drains shall be inspected, and damage shall be repaired to prevent concrete from leaking into the ducts.

The grout fittings and the ducts shall be adequately protected from collapse and other damage. Grout openings, vents and drains must be securely anchored to prevent displacement during concrete placing operations.

The ends of the ducts shall be covered to prevent the entry of water or debris after the installation in the forms.

The ducts shall be blown out with compressed, oil-free air immediately after the completion of the concrete pour. The compressed air shall be used to the extent necessary to break up and remove any mortar in the ducts before it hardens.

The ducts shall be blown out clear of any water or debris again prior to the placing of the tendons.

Duct ends at match-cast segment ends shall be fitted with gaskets or other sealing devices to prevent infiltration of epoxy during assembly. A uniform pressure of approximately 0.3 MPa shall be applied to the joint during the epoxy curing period. The pressure may be supplied by stressing a cable to a predetermined level, or by other means.

Segmental section joints of 10 mm to 70 mm wide shall be filled with a sand-cement mortar. Joints over 70 mm wide shall be filled with concrete. The strength of the mortar or concrete used in joints shall be at least equal to the parent concrete. The Contractor shall colour the mortar or concrete to match the parent concrete. The Contractor shall submit a trial mix with a proportion of white Portland cement, if necessary, to the Ministry Representative for review.

The concrete shall be vibrated with particular care at each cast-in-place end block (push-through diaphragm) and post-tensioning anchorage location. This will ensure adequate consolidation in the anchorage zones.

If the post-tensioning steel is installed prior to placing of concrete in the cast in place elements, then the prestressing steel in each tendon shall be pulled immediately after concreting, to ensure that it is free to move and is not bonded by concrete or grout leakage.

416.34 Stressing – The structure shall be stressed only after the concrete has reached its transfer strength as shown on the Drawings. No tendons shall be stressed until authorized by the Ministry Representative.

Tendons shall be stressed from the end as shown on the Drawings.

The supports under the segmental sections shall be able to accommodate horizontal movements caused by the stressing.

The multiple strands composing a tendon in a common duct shall be tensioned simultaneously. Tendons shall be tensioned in sequence as specified in the reviewed procedure. The tendon force at lock-off shall be as shown in the procedure.

Before commencing the stressing operation an initial tensioning force shall be applied to each tendon to equalize stresses and to eliminate slack. This force shall not exceed 20% of maximum jacking force.

After the initial tension forces have been applied, reference points shall be established for each tendon from which elongation by final tensioning forces will be measured.

The specified tensioning force shall be maintained steady in the strands until they are anchored. The sequence of tensioning shall be such that no undue tension stresses occur in the concrete near the ends of the members.

The tensioning force shall be determined by the calibrated gauge and by the elongation of the strands. When there is a difference of over 5% between the two means of calculation, the cause of the discrepancy shall be ascertained and corrected.

Tendons shall not be subjected to jacking forces in excess of 85% of their specified ultimate tensile strength. Any tendon subjected to such forces will be rejected.

The required tendon elongation and jacking force shall be computed, based on the required prestressing force and the estimated prestress losses.

The Contractor shall mark the ends of each strand adjacent to the chuck wedge after stressing is complete.

The safety of the stressing procedure shall be in accordance with the Occupational Health and Safety Regulation.

Stressing records shall be kept by the Contractor. Stressing records shall clearly identify the tendons that were stressed, the sequence, the final jacking loads, the tendon elongations, the times, the temperatures and the dates. Stressing records shall be signed by the person supervising the stressing and submitted to the Ministry Representative.

416.35 Grouting – The temperature of grout shall be controlled by water temperature. Grout shall not be warmer than 30°C or colder than 5°C at the pump. The grout shall be mixed for a minimum of 2 minutes. The materials shall be batched in this order: water, cement and admixture. The admixture shall be added in accordance with the manufacturer's instructions.

The grout shall be mixed to a uniform consistency of thick cream without lumps or undispersed cement. The grout shall be agitated continuously and shall be recirculated when not injecting from the line.

Water shall not be added to the grout to increase the flowability that was decreased by delayed use of the grout. The grout shall be discarded when flowability has so decreased it cannot be pumped satisfactorily.

The time between the tensioning and grouting of a tendon shall not exceed 4 days.

Cored ducts with concrete walls shall be flushed with clean water to ensure that concrete is thoroughly wetted prior to grout injection. The ducts for external tendons shall be checked for leakage and blockage by flushing with clean water under pressure followed by oil-free compressed air immediately prior to grouting. Metal sheaths for internal ducts shall be flushed to clean out foreign materials, or oil free compressed air shall be used to check for blockages.

All grout and high point vent openings shall be open when grouting starts. Grout shall be allowed to flow from the first vent until any residual flushing water or entrapped air has been removed. The vent shall be capped or otherwise closed

at that time. The remaining vents shall be closed in sequence in the same manner.

The pumping pressure at the tendon inlets shall be at least 0.7 MPa and shall not exceed 1.5 MPa.

If the pumping pressure exceeds the maximum allowable, grout may be injected at any grouted vent, as long as a one-way flow of grout is maintained. If this procedure is used the vents that are to be used for injection shall be fitted with positive shutoffs.

The grout shall be pumped through the ducts and continuously wasted at the outlet pipes until no visible slugs of water or air are ejected. The efflux rate of the ejected grout shall not be less than the influx rate of the injected grout.

To ensure that the tendons remain filled with grout, the outlets shall be closed, and the pressure held for 10 seconds. The grouting ends shall then be plugged. Plugs, caps or valves shall not be removed or opened until the grout has set. Alternatively, the outlets and inlets shall be kept under a 1000 mm head of hydrostatic pressure. The 1000 mm grout head shall be maintained until the grout has hardened.

If the grouting operation is interrupted or it cannot be completed due to mechanical failure or some other situation, then the grout shall be immediately flushed out of the ducts with water when one-way flow of grout cannot be maintained. Standby water flushing equipment, with its own source of power, shall be available in addition to the grouting equipment. It shall be capable of developing a pressure of 2 MPa, and sufficient capacity to flush out any partially grouted enclosures. After flushing, the ducts shall be blown out with oil-free compressed air and then checked for blockages. The grouting procedure shall be repeated if there are no blockages.

The ducts shall be completely filled with grout. The presence of ungrouted lengths of duct shall be cause for nonacceptance of the members. Members with partially ungrouted duct(s) shall be replaced or repaired to the satisfaction of the Ministry at the Contractor's expense.

Great care shall be taken to prevent blockages of the ducts due to grouting of nearby ducts or other causes.

All costs for investigations and for remedial measures due to blockages shall be at the Contractor's expense.

The grouting shall not be done when air temperatures are above 32°C or below 5°C, nor when freezing temperatures are predicted within the next 24 hours.

The temperature of the concrete shall be 2°C or higher from the time of grouting until site-cured cubes of grout reach a compressive strength of 6 MPa.

The grouted ducts may be subject to non-destructive testing inspection by the Ministry to ensure that all ducts are fully grouted. The cost of the inspection will be borne by the Ministry if the inspection indicates that ducts are completely filled with grout. Otherwise, the costs of inspection and the cost of reinspection shall borne by the Contractor.

416.36 Cropping – The cropping of post-tensioning strand shall be by a cold-cutting procedure. It shall not take place until authorized by the Ministry. The strand marks placed at end of stressing shall be inspected to ensure that no slippage of strand has occurred prior to cropping. Under no circumstances shall heat be applied to post-tensioning strand from cropping or other operations. The cropped strand shall be provided with a minimum concrete cover of 25 mm.

416.37 Anchorage Recesses – The anchorage recesses shall be cleaned and sandblasted after completion of post-tensioning and grouting. The surfaces shall be coated with an epoxy bonding agent conforming to ASTM C881, Type V, Grade 2. The Class of bonding agent shall be appropriate for the temperature at time of application. The bonding agent shall be applied in accordance with the manufacturer's applications. Wire mesh acceptable to the Ministry representative shall be placed before concreting in recesses larger than 300 mm square.

The Contractor shall fill the recesses flush with adjoining surfaces. All parts of anchorages shall have a minimum cover of 25 mm. All recesses exposed to view shall be filled with concrete coloured to match the parent concrete. If requested, the Contractor shall submit a trial mix with a proportion of white Portland cement to the Ministry Representative for review.

PAYMENT

416.90 Payment – Payment for Manufacture and Installation of Post-Tensioned Concrete Members will be made at the Lump Sum Price bid. Payment shall include supply and pumping of grout into the post-tensioning ducts, patching of the anchorage recesses, supply and placing of elastomeric pads and strips and grout between stringers, and supplying, placing and tensioning of strands. Payment shall also include splicing of precast segments (if required) and the supply of all material and labour.

SECTION 418

SILANE TREATMENT OF NEW CONCRETE BRIDGE SURFACES

418.01 Scope – This Section describes the treatment of new concrete bridge surfaces with a silane protective solution. The Specification describes surface preparation and application.

Surfaces to be treated shall be as specified on the Drawings or in the Special Provisions.

418.02 Silane Solution – The silane solution shall be carried in a solution to meet the Canadian Environmental Protection Act, Volatile Organic Compound (VOC) Limits for Architectural Coatings Regulations, SOR/2008-264.

418.02.01 Delivery and Storage of Silane Solution – The silane solution shall be delivered to the Site in unopened, sealed containers with original seals intact, also with a manufacturer's label identifying the product (including a batch number). The containers holding silane solution shall be stored in dry, enclosed structures to protect the contents from misuse, weather and excessive heat.

418.02.02 Quality Assurance – Only silane solutions listed in the Recognized Products List shall be permitted for use in the Work.

The Recognized Products List may be found on-line at:

> https://www2.gov.bc.ca/assets/gov/driving-and-transportation/transportation-infrastructure/engineering-standards-and-guidelines/recognized-products-list/recognized_products_list.pdf?forcedownload=true

Certification shall be required for each batch shipment of silane solution, prior to its use.

418.03 Contractor's Qualification – The applicator shall have a minimum of three years of successful and documented experience in application of liquid water repellents to concrete or masonry substrates, or else the applicator must provide documentation certifying that training approved by the manufacturer for applying silane solution was received. Certification shall be provided to the Ministry Representative upon request.

418.04 Site Conditions

418.04.01 Substrate Condition – Prior to silane treatment:

(a) New concrete shall be cured for a minimum of 28 days.

(b) All concrete surfaces shall be fully dry.

418.04.02 Weather Limitations – Application of silane solution shall be undertaken only during periods when rain or snow has not occurred during the previous two days and no rain or snow is forecast by the local weather office for the next two days.

Application of silane solution shall proceed only when the concrete surface temperature is above 4°C and below 32°C, and windspeed is less than 24 km/hr.

418.05 Surface Preparation – All concrete surfaces prepared for treatment shall be thoroughly cleaned prior to application of the silane solution. The method of cleaning shall be capable of removing loose concrete, curing compounds, laitance, dirt, dust, salt, oil, asphalt or other foreign materials. The cleaning process shall expose concrete pores and capillaries to readily accommodate silane penetration.

The cleaning process shall not damage the concrete surface, remove or alter the existing surface finish, or expose the coarse aggregate of the concrete. The method of cleaning shall be performed in such a manner as to provide a uniform appearing surface colour.

418.05.01 Pre-Surface Cleaning – If necessary, pre-surface cleaning preparations may include the use of solvent and hand tools to remove materials detrimental to the silane treatment of the concrete surface.

418.05.02 Surface Cleaning – Suitable methods to clean the concrete surface are:

(a) Sand blasting

(b) Steel shotblasting

(c) Hydro blasting

(d) Steam cleaning

(e) Water pressure washing using a minimum of 3.5 MPa (500 psi) nozzle pressure.

The cleaning method used shall be capable of cleaning concrete surfaces, as specified.

418.06 Silane Application – Silane application shall proceed only if the condition of the cleaned concrete surface is accepted by the Ministry Representative.

418.06.01 Surface Condition – All concrete surfaces shall be fully dry prior to silane solution application. Concrete surfaces previously steam cleaned or water washed must be allowed to dry prior to application of silane solution for:

(a) Minimum one day during windy or sunny days when air temperatures are over 20°C.

(b) Minimum two days during calm or cold days when air temperatures are between 4°C to 20°C.

418.06.02 Silane Solution – The silane solution shall be used as supplied by the manufacturer. It shall not be diluted or

altered in any way. Water shall not be allowed to contaminate the silane solution.

418.06.03 Application Equipment – The silane solution shall be sprayed on the concrete surface by using low pressure 0.1 – 0.2 MPa (15 – 30 psi) positive displacement airless spray equipment. Suitable equipment is a pressurized garden sprayer or drum mounted pump. Large deck areas over 400 m^2 shall be sprayed using a spraybar type unit.

All spray equipment shall be fitted with fan spray nozzles and the spray adjusted to a "wet spray" condition (i.e., minimum atomization of silane solution).

(a) Cleaning of Application Equipment – All application equipment must be clean and dry prior to use. If equipment is wet, it shall be rinsed with a small amount of methyl hydrate (methanol) prior to commencement of the actual application.

418.06.04 Rate of Application

(a) Surfaces Subject to Traffic Wear – Bridge decks and other surfaces subject to traffic wear shall be treated with silane solution at a rate of 0.40 L/m^2.

(b) Surfaces not Subject to Traffic Wear – Surfaces not subject to traffic wear (parapets, pier caps, sidewalks, etc.) shall be treated with silane solution at a rate of 0.23 L/m^2.

418.06.05 Method of Application – Silane solution shall be applied at a coverage rate according to this Specification. Horizontal surfaces shall be uniformly sprayed to saturate/flood the surface. Vertical surfaces should be treated from the bottom up and uniformly sprayed to saturate the surface.

If it is not possible to complete the silane treatment in one application, the boundary between the treated and untreated areas shall be clearly marked.

418.06.06 Curing – Following the application, the surface of silane treated concrete shall be cured by mist spraying with water after:

(a) 4-6 hrs, during days with temperatures between 25°C to 32°C;

(b) 6-12 hrs, during days with temperatures between 15°C to 25°C; or

(c) 12-24 hrs, during days with temperatures between 4°C to 15°C.

The traffic may be allowed on silane treated concrete one hour after the water wetting operation.

418.07 Safety and Health Hazards – Personnel exposed to silane solution and methyl hydrate shall wear personal protective equipment such as coveralls, goggles, rubber gloves and proper respirators.

No smoking, open flames, sources of heat or ignition shall be permitted during the construction operation.

Special precautions must be taken during construction to prevent any over spray or spillage of materials on plants and vegetation or into fish-habitat water bodies and work shall be in accordance with SS 165, Protection of the Environment.

Manufacturer's material safety data and handling instructions, including disposal of containers, shall be followed.

418.08 Measurement and Payment – Payment for the concrete deck sealant will be made at the Unit Price bid per square metre for the preparation of the concrete surfaces and the supply, application and curing of the deck sealant.

SECTION 419

CONCRETE BRIDGE DECK WATERPROOFING SYSTEM PREFABRICATED MEMBRANE

DESCRIPTION

419.01 Scope – This Section describes the protection of concrete bridge decks with Rubberized Asphalt Prefabricated Membrane (Prefabricated Membrane) and an overlay of hot applied asphalt pavement.

The area to be covered with Prefabricated Membrane shall be as shown on the Drawings or as described in the Special Provisions.

This Section describes supply, installation, materials, surface preparation, application, and other related requirements.

NOTE: Prefabricated Membrane shall not be used on Ministry concrete bridge decks contaminated with acid-soluble chloride ion (Cl^-) in excess of 0.10% by mass of cement.

419.02 Applicator's Qualification – Only applicator(s) trained and certified by the Prefabricated Membrane supplier/manufacturer, whose product is used in the Work, will be permitted to install waterproofing membrane and accessories. Certification documentation shall be provided to the Ministry Representative upon request.

Special Provisions may identify alternate acceptable documentation for remote locations and special requirements.

419.03 Quality Control – Installation and quality of the Prefabricated Membrane shall be inspected and corrected by the Contractor in accordance with this SS 419 and the supplier/manufacturer's requirements. If there is a discrepancy between the two, the more stringent requirement shall apply. In cases where it may be unclear as to which requirement is more stringent, the Ministry Representative will have the final say.

The Contractor shall provide a copy of the supplier/manufacturer's quality control certification for each lot of material prior to its use in the Works.

419.04 Quality Assurance – The Ministry will implement a quality assurance program by auditing the Contractor's quality control program and by inspection at its discretion.

The Contractor shall notify the Ministry Representative at least 48 hours before installation of the Prefabricated Membrane. Inspection by the Ministry shall not relieve the Contractor from obligation to perform the work in accordance with the Contract.

MATERIALS

419.11 General – Materials shall be provided as follows.

419.11.01 Prefabricated Membrane

(a) Selection – The Contractor shall use a Prefabricated Membrane selected from the Ministry's Recognized Products List (RPL) under the category "Prefabricated Membrane for Concrete Decks".

(b) Supplier – The Prefabricated Membrane shall be supplied by the supplier/manufacturer of the prefabricated membrane as a complete package.

(c) Package – The Prefabricated Membrane package shall be comprised of the following materials: primer, prefabricated membrane detailing strip, full width prefabricated membrane and mastic.

(d) Information Requirements – Two weeks prior to the application of the Prefabricated Membrane, the Contractor shall provide the Ministry Representative with a list of the supplier/manufacturer's materials for the selected Prefabricated Membrane and also provide detailed specifications, installation procedures, technical data sheets, safety data sheets and quality control procedures.

419.11.02 Primer – The Primer used in conjunction with the Prefabricated Membrane shall be a rubberized asphalt or polymer dispersed in organic solvent with minimum solids of 45% (by mass), and maximum drying time of one hour.

419.11.03 Prefabricated Membrane Detailing Strip – Prefabricated Membrane Detailing Strip (Membrane Detailing Strip), shall be a self-adhering internally reinforced sheet of rubberized asphalt and shall have thickness between 1.50 and 1.75 mm (60 to 70 mils) and shall be supplied in rolls of 300 mm width.

419.11.04 Full Width Prefabricated Membrane – Full Width Prefabricated Membrane shall be a self-adhering internally reinforced sheet of rubberized asphalt. Prefabricated Membrane shall have a minimum thickness of 1.5 mm (60 mils) and shall be supplied in rolls of width between 900 mm and 1500 mm.

419.11.05 Mastic – Mastic shall be a single component material, consisting of rubberized asphalt, organic solvent and fillers, supplied in caulking tubes.

419.12 Delivery and Storage – Prefabricated Membrane and accessory materials shall be delivered to the Site in unopened packages and containers with manufacturer's label identifying the product and a batch number.

The packages and containers shall be stored in a dry secured place protected from weather and excessive heat. Specific manufacturer's instructions for handling and storage shall be followed.

CONSTRUCTION

419.31 Traffic Restrictions – All traffic, other than construction equipment directly associated with the installation of the Prefabricated Membrane and placement of asphalt pavement shall be prevented from using areas of the deck that have received surface preparation until paving of that area is completed.

419.32 Job Site Conditions – In addition to the supplier/manufacturer's specifications, the following conditions shall apply.

419.32.01 Weather Limitations – Application of Primer and installation of Prefabricated Membrane and pavement overlay shall be undertaken only during stable weather when precipitation has not occurred during the previous 48 hours and is not imminent and when the minimum ambient and concrete surface temperature is at least 5°C and rising**.**

419.32.02 Substrate Conditions – Prior to the application of Prefabricated Membrane, new concrete shall be cured for a minimum of 28 days or as otherwise required by the Ministry Representative to ensure the extent of substrate moisture from the hydration process is not trapped beneath the Prefabricated Membrane nor restricting the penetration of the Primer.

419.33 Surface Preparation – All concrete surfaces for application of a Prefabricated Membrane shall be sound, smoothed to a profile of less than 2 mm, clean, and free from any contamination.

419.33.01 Preparation Conditions

(a) All defects in concrete surfaces, cracks, delamination and spalled areas shall be repaired with a suitable patching material. All patching shall be cured to the acceptance of the Ministry Representative.

(b) All protrusions shall be removed, and all edges shall be rounded to a minimum radius of 10 mm.

(c) Concrete bridge decks and concrete-filled shear keys between prestressed concrete box stringers shall have smooth finish. Broom finishes are not acceptable.

(d) All loose aggregates, laitance, dust, soil, oil and other debris and contaminants shall be removed from the concrete surface by one or more of a combination of the following methods:

(i) high pressure clean dry air sweeping;

(ii) sand blasting;

(iii) steel shot blasting;

(iv) hydro blasting using a minimum of 17.5 MPa (2500 psi) nozzle pressure; and

(v) steam cleaning.

The method used shall be capable of removing concrete laitance and cleaning concrete surfaces as specified to meet the Prefabricated Membrane manufacturer's requirements for a clean and smooth surface.

419.34 Priming – All concrete deck surfaces that are to receive Prefabricated Membrane must be treated with a Primer. Primer shall be used as supplied by the manufacturer. It shall not be diluted or altered in any way.

419.34.01 Concrete Surfaces – All concrete surfaces shall be clean and fully dry prior to application of Primer. Concrete surfaces previously steam cleaned, or water washed must be allowed to dry prior to application of Primer for:

(a) minimum 24 hours during windy or sunny days when air temperatures are over 20°C.

(b) minimum 48 hours during calm or cold days when air temperatures are between 5° to 20°C.

419.34.02 Primer Application – Primer application shall proceed only if the condition of cleaned and repaired concrete surfaces is approved by the Ministry Representative.

All concrete surfaces to be covered with Prefabricated Membrane shall be uniformly treated with Primer at the rate recommended by the supplier/manufacturer (typical rates are 0.15 to 0.25 L/m², depending on the porosity of the concrete).

The Primer shall be uniformly applied to the concrete deck surface by roller or brush.

419.34.03 Primer Curing – Only as much surface as will be covered by Prefabricated Membrane in the workday shall be primed. Primed concrete surfaces not covered by Prefabricated Membrane within the workday must be reprimed immediately prior to Prefabricated Membrane installation.

All primed areas must be tack free prior to application of the Prefabricated Membrane.

419.35 Prefabricated Membrane Installation

419.35.01 Detail Work – Extra attention must be given to all detail work which shall be performed prior to final installation of the full-width membrane.

The work shall be completed in accordance with SS Drawings SP419-01 through SP419-08, other related Special Provisions and as follows:

(a) **Curb or Parapet** – inside corners at curbs or parapets shall be covered using a Membrane Detailing Strip.

The uppermost edge of the Membrane Detailing Strip shall be placed and extended up the face of curb or parapet to 6 mm (1/4") below the finished height of the asphalt pavement overlay and terminated in a 12 mm deep saw cut in the face. The Membrane Detailing Strip shall be installed in such a manner to ensure it is smooth and that it adheres strongly to the face of curb or parapet, and bridge deck without any visible defects ("tenting", "fish mouth" or "bubbles"). The first sheet of the full width of the Prefabricated Membrane shall then be applied as close as possible to the curb or parapet.

(b) **Bridge Deck Edge** – on bridge decks without curbs or parapets, the full width Prefabricated Membrane or Membrane Detailing Strip shall terminate 6 mm (1/4") from the outer edges of the deck. The Prefabricated Membrane shall be heated at the edge of the deck with light flame from a torch or with a hot air gun, followed by pressing the Prefabricated Membrane down with a roller.

(c) **Drains** – areas around drains, basins and any other protrusions shall be double covered with Prefabricated Membrane. First, flashing pieces of Membrane Detailing Strip are applied, followed by the application of the full width Prefabricated Membrane. The Prefabricated Membrane shall be installed in such a way as not to obstruct openings of drains and basins.

(d) **Weep Holes** – do not require a Membrane Detailing Strip. The diameter of weep holes should be 25 mm (1"). The Prefabricated Membrane around weep holes shall be heated with light flame from a torch or hot air gun, and pressed down to expose the opening of the hole. Where the Drawings do not indicate weep holes, the Ministry Representative may identify locations for weep holes at approximately every 3 m along the edge of the curb or deck and the work to install these additional weep holes will be paid for as Extra Work.

(e) **Shear Keys, Joints and Cracks** – All shear keys, joints with a gap less than 6 mm , and all filled cracks wider than 3 mm shall be covered with a 300 mm wide Membrane Detailing Strip prior to application of a full width Prefabricated Membrane. Joints wider than 6 mm shall be treated as a standard expansion joint and Prefabricated Membrane shall be terminated at the expansion joint assembly.

(f) **Bridge Ends** – A 150 mm wide Membrane Detailing Strip shall be installed over the joint between the existing pavement and the bridge deck. At each end of the bridge deck, a full-width Prefabricated Membrane shall extend at least 900 mm over the paved bridge approaches. In the case of Prefabricated Membrane application on an existing structure with gravel approaches, the Membrane Detailing Strip of Prefabricated Membrane shall be folded 150 mm over the deck edge and covered with a full width Prefabricated Membrane. See SS Drawings SP419-05 to SP419-07.

419.35.02 Installation of Prefabricated Membrane – The full width Prefabricated Membrane shall be applied in a longitudinal direction from the low point to the high point so that laps shed water. The area along the lower edge of the deck shall be covered first.

The full width Prefabricated Membrane shall be laid flat, avoiding wrinkles and entrapped air. Each succeeding sheet of Prefabricated Membrane shall be applied overlapping the side laps a minimum 65 mm and the end laps a minimum 150 mm. The entire surface of the Prefabricated Membrane, especially laps, and all terminations shall then be rolled with sufficient pressure to affect a bond between the Prefabricated Membrane and the concrete surface.

419.35.03 Mastic Application – All Prefabricated Membrane terminations shall be sealed with a trowelled application of mastic 12 to 25 mm wide and 1.5 to 3.0 mm thick. Mastic shall be applied along the edge of the Prefabricated Membrane at the face of the curb or parapet, along both sides of the deck joints, around drains and weep holes and at termination of the Prefabricated Membrane on both ends of bridge approaches.

At the end of each working day, or when application is interrupted by a change in construction scheduling or weather, all Prefabricated Membrane terminations shall be sealed with the mastic.

419.35.04 Inspection and Repairs – Prior to paving over the Prefabricated Membrane, any defective areas of the Prefabricated Membrane shall be repaired to the satisfaction of the Ministry Representative, including:

(a) tears, damaged areas and inadequately lapped seams shall be patched;

(b) fish mouth shall be slit and repaired;

(c) large blisters shall be punctured and repaired; and

(d) if water migrates underneath the installed Prefabricated Membrane, then the Prefabricated Membrane covering the affected area shall be removed and replaced.

419.35.05 Traffic on Prefabricated Membrane – Traffic shall not be permitted on the Prefabricated Membrane prior to installation of the first lift of asphalt pavement with the exception of vehicles and equipment engaged in installing the asphalt pavement. Only vehicles or equipment provided with pneumatic tires or with flat tracks with rubber bonded pads, shall be permitted on the Prefabricated Membrane.

Any turning or sudden braking on the Prefabricated Membrane shall be avoided. Any damage to the Prefabricated Membrane resulting from the movement of

vehicles or equipment shall be repaired at the Contractor's expense.

419.36 Paving – Asphalt overlay construction shall proceed only when the final condition of the installed waterproofing Prefabricated Membrane is accepted by the Ministry Representative.

Prior to commencing asphalt pavement construction, the weight of equipment, including loaded trucks, paver and rollers, and their maximum combined weight must be accepted by the Ministry's responsible Bridge Engineer. Only this accepted equipment and equipment combination shall be permitted on the bridge structure at the time of construction.

Asphalt materials, mix design, mix production and hot mix asphalt pavement construction shall meet the requirements of SS 502 and SS 952. Special Provisions will identify any changes to SS 502 bonus/penalty requirements for the bridge application.

419.36.01 Asphalt Mix and Pavement Design – Asphalt mix shall meet the requirements for Pavement Class 1, Medium Mix and appropriate asphalt binder. The asphalt pavement mix shall be paver laid and compacted in two nominal 50 mm lifts.

419.36.02 Placing the Asphalt Mix – In addition to standard paving practice:

(a) paving shall only proceed over dry Prefabricated Membrane;

(a) no asphalt primer shall contaminate the Prefabricated Membrane surface;

(b) paving operation shall commence as soon as possible after the waterproofing Prefabricated Membrane is installed;

(c) paving shall proceed in the direction of the end laps in the Prefabricated Membrane (i.e., from overlap to underlap); and where more than one width of pavement is to be laid the first width shall be laid starting at the lower edge transversely (i.e. on super elevated bridges the first mat of pavement is to be started at the lower edge of the deck to avoid shifting of the mat on the membrane).

(d) the Contractor shall take due precautions to avoid damage to the waterproofing Prefabricated Membrane during the paving operations by the paver and hauling trucks;

(e) the paver shall move continuously and at constant speed avoiding any unnecessary stopping or turning;

(f) hauling trucks shall maintain a steady supply of asphalt mix to the paver;

(g) excessive accumulation of asphalt mix in the paver hopper and auger shall be avoided;

(h) unloading of the asphalt mix from the hauling trucks onto the Prefabricated Membrane surface will not be permitted;

(i) paver screed burners shall be set in such a way as to avoid any damage to the Prefabricated Membrane from excessive heat;

(j) the path in front of the screed side end plate must be kept continuously free from spilled asphalt mix. The bottom of the side end plate shall slide freely, otherwise the entrapped asphalt mix could damage the underlying Prefabricated Membrane;

(k) screed adjustments during paving shall be made in such a way as to achieve a specified thickness of mat, surface texture, smoothness and designed pavement geometrics;

(l) the bridge deck joints, drains and other openings (except weep holes) shall be protected to avoid intrusion of asphalt during the paving operation; and

(m) during a "one lane at a time" construction method, as on bridges partially open to traffic, the constructed pavement mat shall terminate not less than 150 mm from the edge of installed Prefabricated Membrane in order to provide a sufficient area for lapping of the next sheet of the Prefabricated Membrane.

419.36.03 Asphalt Pavement Compaction – In addition to standard compacting practice:

(a) drum vibration mode must not be used during compaction.

(b) steel drum shall run as close as possible to the outer edge, curb or parapet. Outer edge, curb or parapet areas that are inaccessible by the steel drum shall be compacted by a plate compactor.

(c) compaction along the end asphalt plug fillet, expansion joints or concrete dams shall be performed transversely with a steel drum or plate compactor if access by the steel drum is not practicable.

(d) sudden stops or sharp turns by rollers shall be avoided.

419.36.04 Hand Work – Placing of asphalt mix in areas inaccessible to large paving equipment shall be done in accordance with SS 502.

419.37 Protection of the Environment – The Contractor shall comply with SS 165, appropriate environmental requirements of Fisheries and Oceans Canada, the Ministry of Environment, and any related Municipal Regulations.

Special precautions shall be taken during the construction operations to prevent spillage of materials into water bodies.

419.38 Safety and Health Hazards – The Contractor shall obtain all relevant Safety and Material Data information and appropriate handling instructions on Prefabricated

Membrane and accessory materials from the Supplier/Manufacturer to comply with required health and safety regulations.

The Contractor shall comply with the Workers Compensation Act, the Occupational Health and Safety Regulation and manufacturer's requirements for handling and installing Prefabricated Membrane and accessory materials as these materials have flammable, toxic, hot and other potentially harmful properties to workers and the public. The Contractor shall ensure personnel wear and use protective equipment.

MEASUREMENT

419.80 General – Prefabricated Membrane will be measured in square metres installed. The area measurement for payment will be based on the actual plan area covered by the Prefabricated Membrane waterproofing system, excluding laps and vertical application areas.

Asphalt pavement will be measured in tonnes in place.

PAYMENT

419.90 Prefabricated Membrane – Payment for Prefabricated Membrane will be made at the Unit Price bid per square metre.

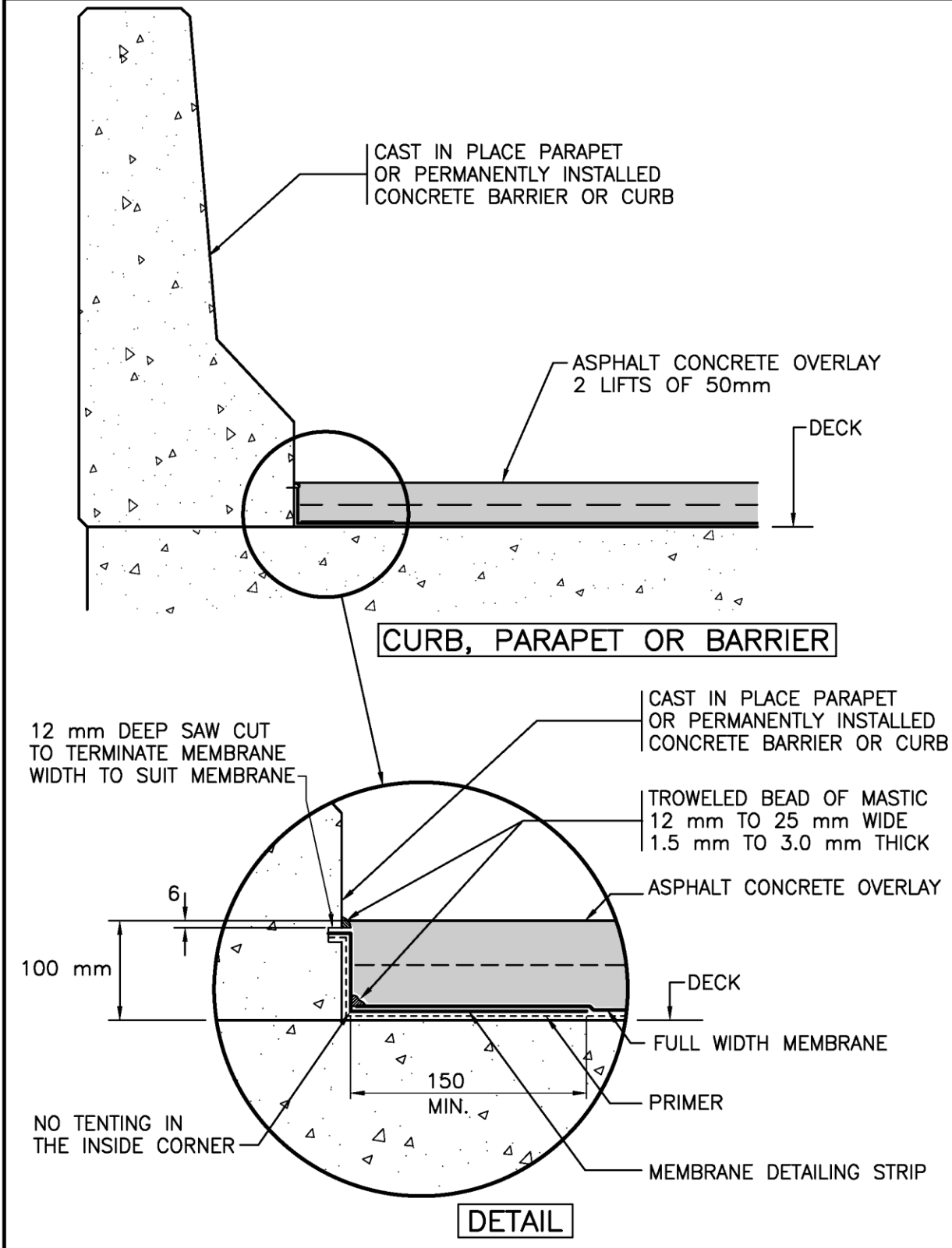
The Unit Price shall include everything done and furnished for the full and complete supply and installation of the Prefabricated Membrane including bridge deck preparation. Payment will include the preparation of the concrete deck surface and the supply and installation of weep holes, Primer, Membrane Detailing Strips, full width Prefabricated Membrane, mastic, and related work for the supply, application and installation of the Prefabricated Membrane.

419.91 Asphalt Pavement – Payment for the Asphalt Pavement and related works to meet the requirements for SS 502 and SS 952 will be made at the Unit Price bid per tonne, subject to any EPS payment adjustments per the Special Provisions and SS 502.

Payment shall include pavement mix design, supply of asphalt cement and asphalt mix aggregates, prime and tack coats, quality control, feeding, blending, mixing, loading, weighing, hauling, dumping, spreading, compacting and finishing of the asphalt pavement overlay and quality control.

BRIDGE DECK – MEMBRANE APPLICATION
TYPICAL DETAIL WORK – CURB, PARAPET OR BARRIER

SP419–01

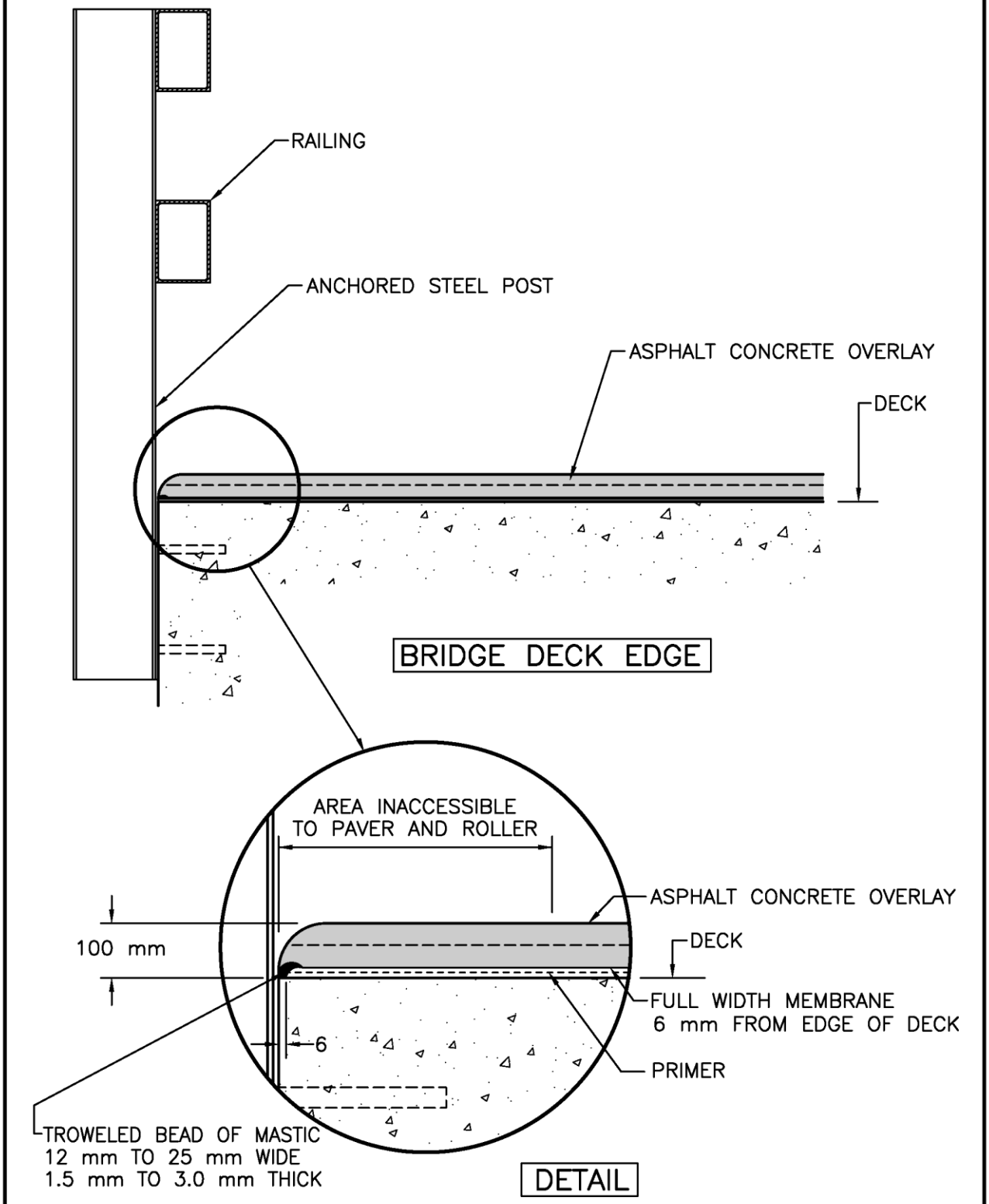


NOT TO SCALE

ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRIDGE DECK – MEMBRANE APPLICATION
TYPICAL DETAIL WORK – BRIDGE DECK EDGE

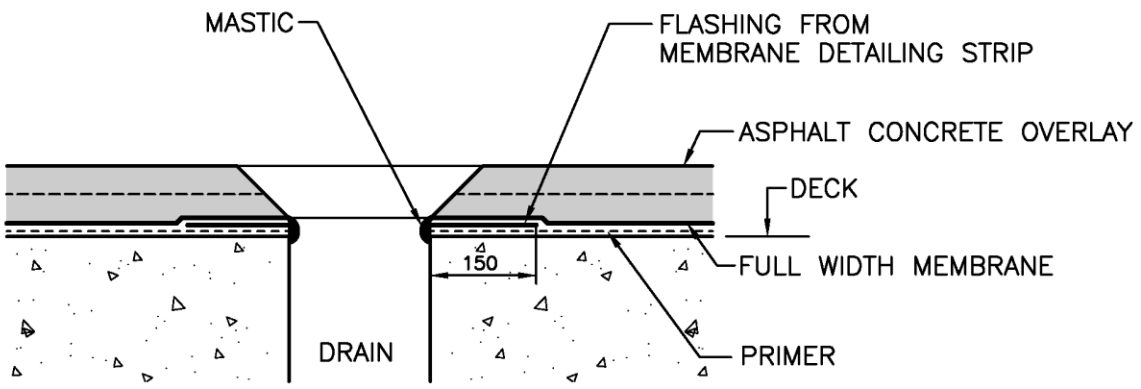
SP419–02



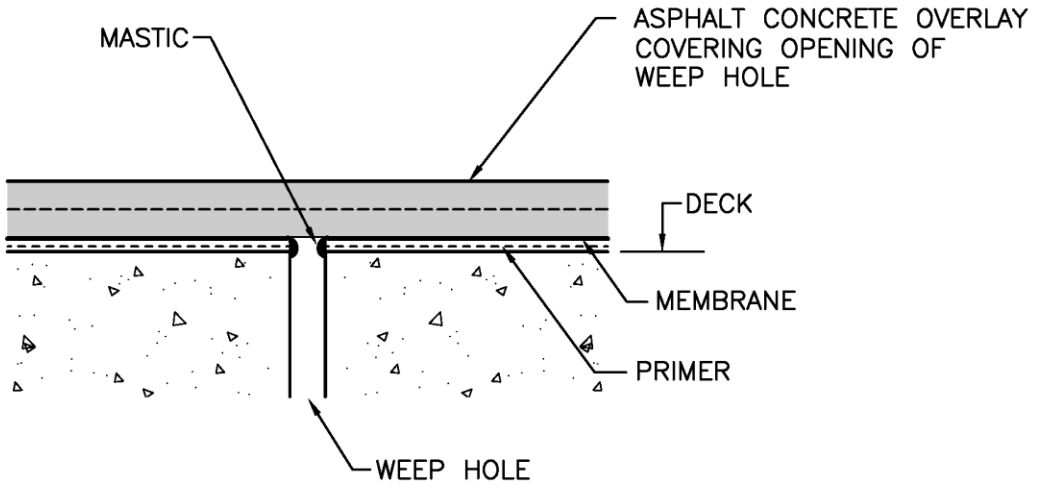
NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRIDGE DECK – MEMBRANE APPLICATION
TYPICAL DETAIL WORK – DRAIN AND WEEP HOLE

SP419–03



DRAIN

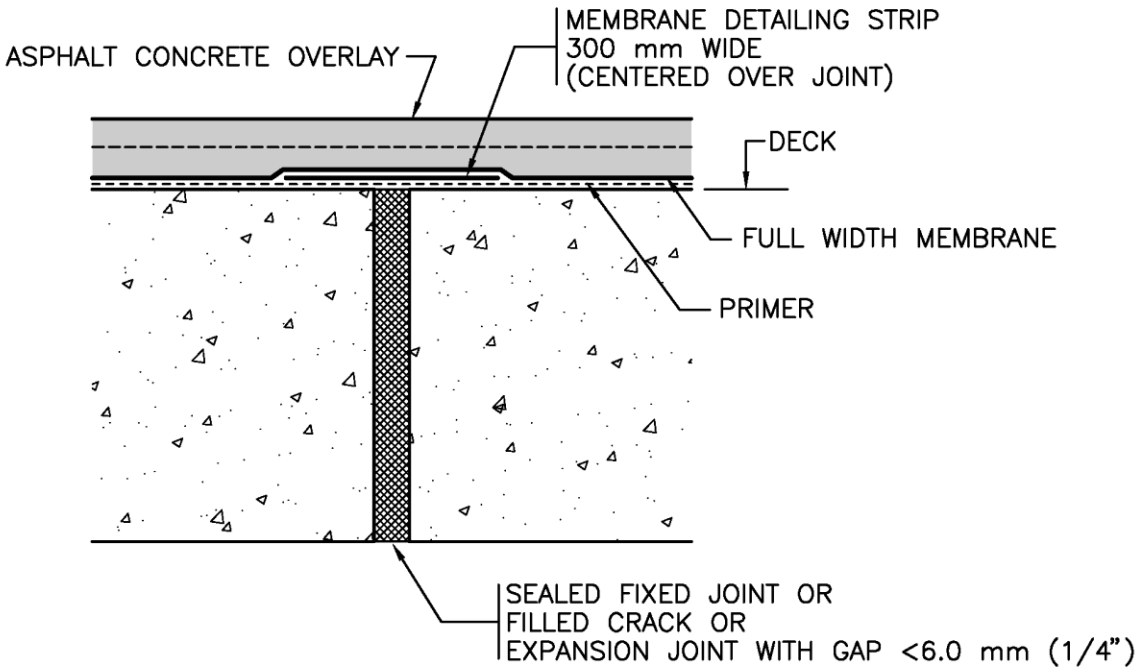


WEEP HOLE

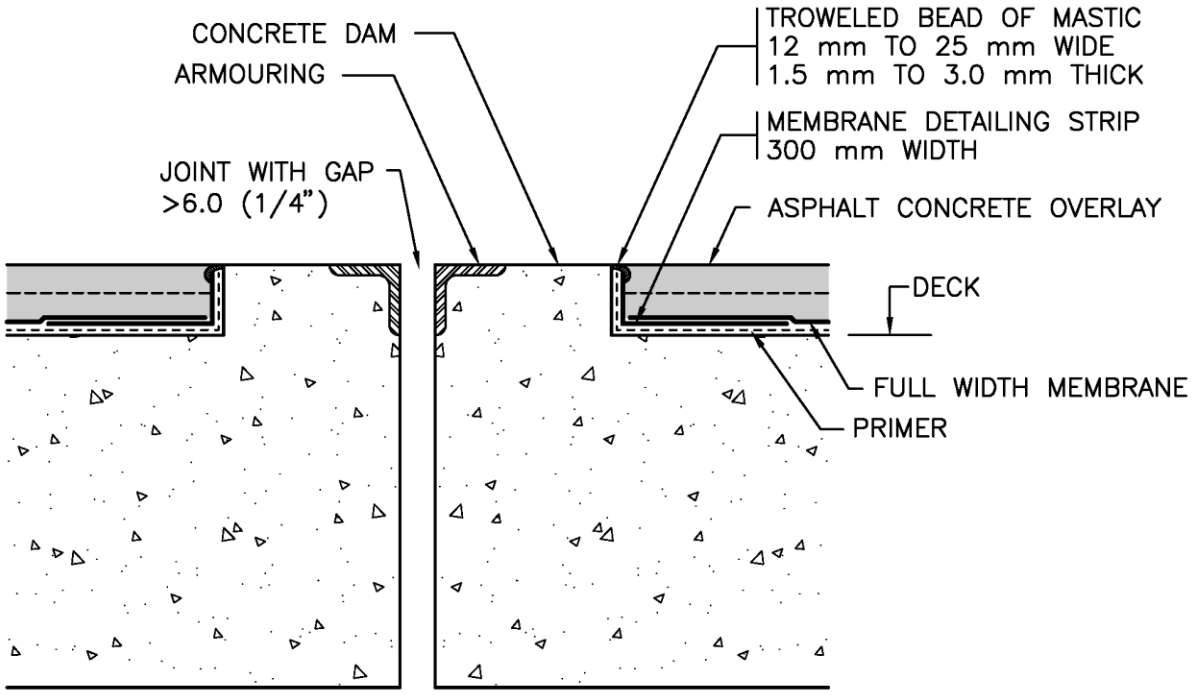
NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRIDGE DECK – MEMBRANE APPLICATION
TYPICAL DETAIL WORK – JOINT AND CRACKS

SP419–04



FIXED JOINT OR CRACK



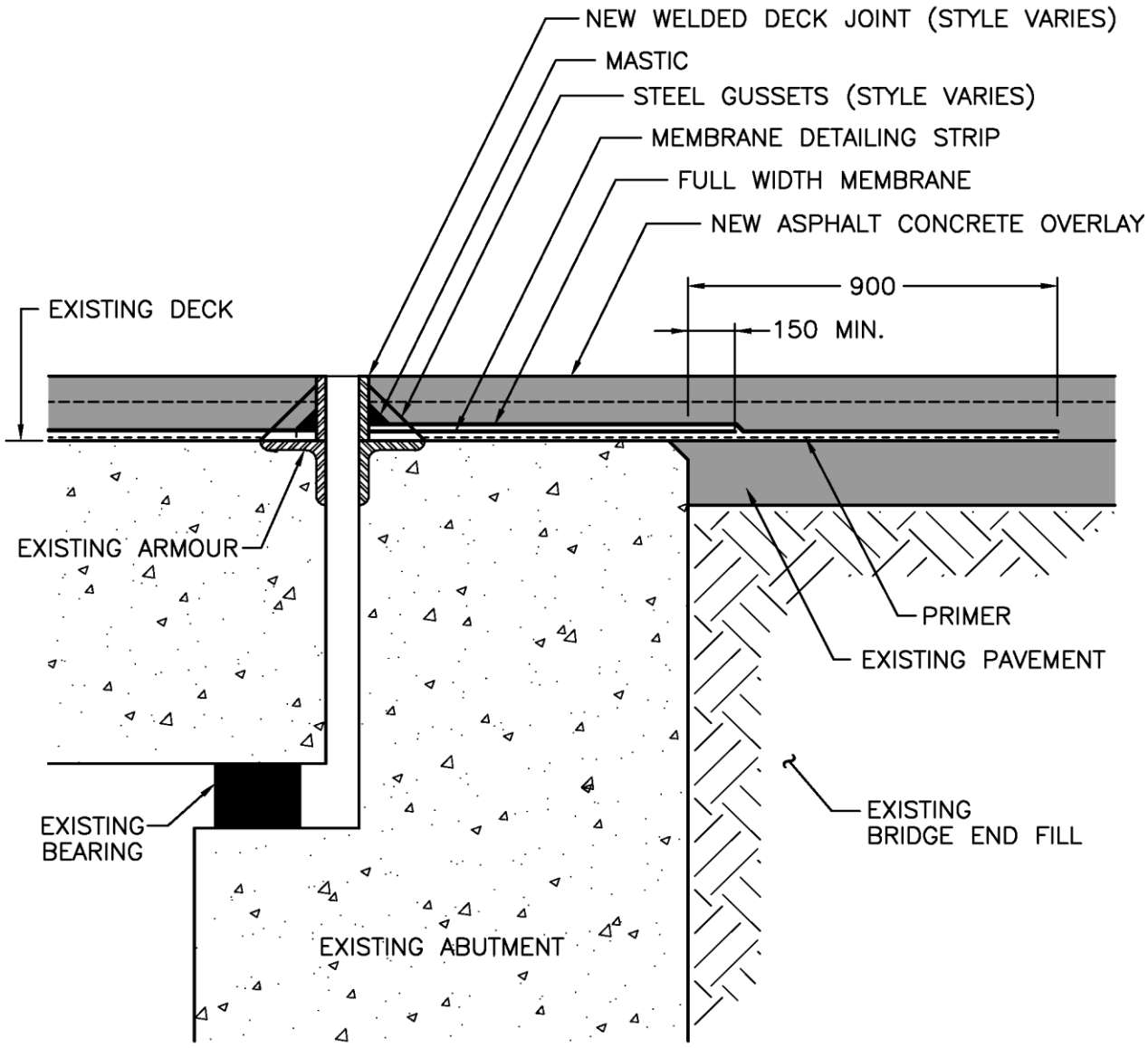
EXPANSION JOINT

NOT TO SCALE

ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRIDGE DECK – MEMBRANE APPLICATION
TYPICAL DETAIL WORK – BRIDGE END

SP419–05

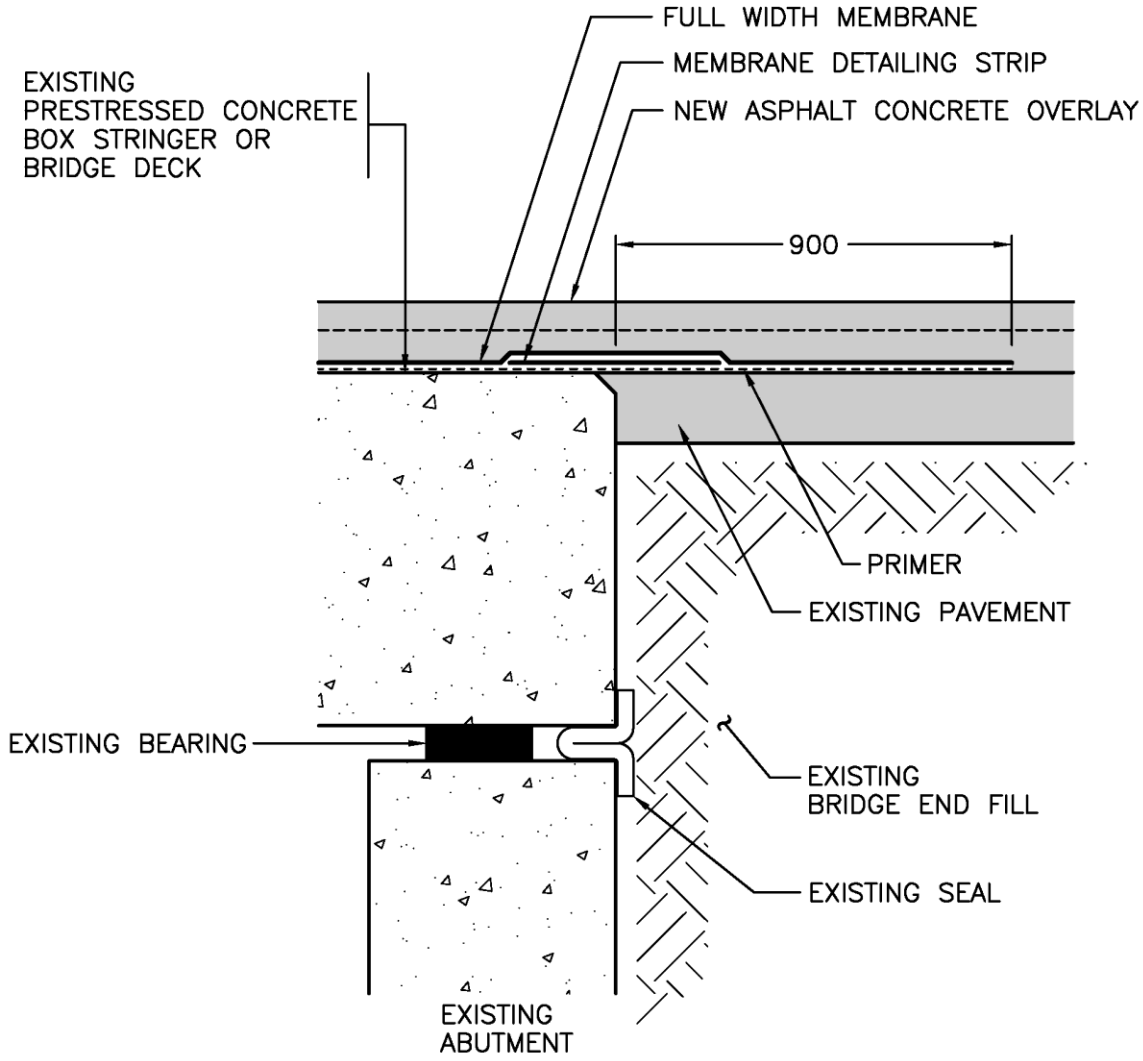


BRIDGE END

(WITH EXPANSION JOINT AND EXISTING PAVED APPROACH)

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRIDGE DECK – MEMBRANE APPLICATION TYPICAL DETAIL WORK – BRIDGE END	SP419–06



BRIDGE END

(EXISTING PAVED APPROACH WITHOUT EXPANSION JOINT)

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED